1405-433 · Statistical Quality Control · Chapter 15

Lot-by-Lot Acceptance Sampling for Attributes

Sampling plans · the OC curve · AQL / LTPD / AOQL · rectifying inspection · MIL STD 105E

Slides covered: **3–9, 12–13, 14–26, 28–29, 39–54**

Summary + all formulas & examples

Contents

11. Quick Review

1 The Acceptance-Sampling Problem slides 3-5

The typical use of acceptance sampling is **lot sentencing** (accept or reject the lot) for receiving inspection. Three aspects are important:

- Its purpose is to **sentence lots, not to estimate lot quality**.
- It does **not control or improve quality** — it only accepts/rejects lots.
- It is an **audit tool** to ensure output conforms to requirements.

Three approaches to lot sentencing

① Accept with **no inspection** · ② **100% inspection** · ③ **acceptance sampling**.

Acceptance sampling is most useful when: testing is destructive; the cost of 100% inspection is very high; 100% inspection isn't technologically feasible; many items + high inspection-error rate; the supplier's history is excellent but process capability is low; or there are serious product-liability risks.

💡 **Advantages vs. disadvantages**

Advantages: less expensive, less handling/damage, applicable to destructive testing, fewer personnel, less inspection error, stronger supplier motivation.
Disadvantages: risk of accepting bad / rejecting good lots; less information generated; requires planning & documentation.

2 Types of Sampling Plans slide 6

Plans are classified by **data type** (variables / attributes) and by the **number of samples** needed for a decision:

Plan	Decision
Single-sampling	Accept or reject from one sample.
Double-sampling	Accept, reject, or take a second sample to decide.
Multiple-sampling	Accept, reject, or take several more samples.
Sequential-sampling	Take units one at a time until accept/reject.

All four can be designed to give **equivalent results**. Selection factors: administrative efficiency, type of information produced, average amount of inspection, and impact on manufacturing flow.

3 Lot Formation & Random Sampling slide 7

- Lots should be **homogeneous**.
- **Larger lots** are preferred over smaller ones.
- Lots should suit the **materials-handling systems** of supplier and consumer.

Units must be selected **at random** and be representative of the lot — the random-sampling concept is extremely important. Sometimes the inspector **stratifies** the lot (divides it into layers/cubes and samples from each) to ensure units come from all locations.

4 Key Quality Levels: AQL, LTPD, AOQL

slides 8–9, 14

Definitions

AQL — Acceptable Quality Level: a “good” quality level considered satisfactory; the poorest process average a consumer would accept. It is a property of the **supplier’s process**, not the plan.
LTPD — Lot Tolerance Percent Defective: a poor-quality level the consumer wants protection against (also called RQL / LQL). Specified by the **consumer**.
AOQL — Average Outgoing Quality Limit: the maximum average defect level after inspection.

Terminology hierarchy

Plan = sample size + accept/reject criteria · **Scheme** = a set of plans (with rules linking lot/sample sizes & 100% inspection) · **System** = a unified collection of schemes.

5 The OC Curve & Risks

slides 12–15

The **operating-characteristic (OC) curve** plots the probability of acceptance P_a against the lot fraction defective p — it shows the plan's **discriminatory power**. For a single-sampling plan (binomial):

Eq 15.2

$$P_a = P\{d \le c\} = \sum_{d=0}^{c} \frac{n!}{d!(n-d)!} p^d (1-p)^{n-d}$$

📊 **Example — n = 89, c = 2**

At p = 0.01: $P_a = P\{d \le 2\}$ = **0.9397**. At p = 0.02 (2% defective): $P_a \approx$ **0.74** — i.e. of 100 such lots we accept ≈ 74 and reject ≈ 26.

⚠ **The two risks**

Type I error (α) = P(reject a good lot) — the producer's risk.

Type II error (β) = P(accept a bad lot) — the consumer's risk.

Effect of n and c: increasing **n** makes the OC curve steeper (more discriminating); changing **c** shifts the curve.

6 Designing a Single-Sampling Plan

slides 16–19

A plan with a specified OC curve can be found with a **Binomial Nomograph** (Fig 15.9). Specify two points:

- **p_1 = AQL**, with $1 - \alpha = P[\text{accept} \mid p = \text{AQL}]$
- **p_2 = LTPD**, with $\beta = P[\text{accept} \mid p = \text{LTPD}]$

Draw line 1 from p_1 to $(1 - \alpha)$ and line 2 from p_2 to β; read **n** and **c** at the intersection.

📊 **Example**

$p_1 = 0.01$, $\alpha = 0.05$, $p_2 = 0.06$, $\beta = 0.10$ → **$n = 120$, $c = 3$**.
"Take a sample of 120; accept the lot if defectives ≤ 3, otherwise reject the entire lot."

7 Rectifying Inspection: AOQ, AOQL, ATI

slides 20–26

A **rectifying inspection** program “corrects” lot quality: rejected lots are sent to **100% inspection (screening)** to remove all defectives. Used to know the average quality reaching the next stage (receiving, in-process, final inspection).

Average Outgoing Quality (AOQ)

The average quality leaving after rectifying inspection (assuming rejected lots are 100% screened & defectives replaced with good units):

Eq 15.4

$$AOQ = \frac{P_a \cdot p \cdot (N - n)}{N} \approx P_a \cdot p \ \ (\text{when } N \gg n)$$

📊 Example — N = 10,000, n = 89, c = 2, p = 0.01

$P_a = 0.9397 \rightarrow AOQ = (0.9397)(0.01)(10{,}000 - 89)/10{,}000 =$ **0.0093** (0.93% defective).

AOQL

The **Average Outgoing Quality Limit** is the **maximum point on the AOQ curve** — the worst possible average quality from the program (e.g. AOQL ≈ 0.0155).

Average Total Inspection (ATI)

Average number of units inspected per lot (varies between n and N):

Eq 15.6

$$ATI = n + (1 - P_a)(N - n)$$

📊 **Example**

$N = 10{,}000$, $n = 89$, $c = 2$, $p = 0.01$, $P_a = 0.9397$ → $ATI = 89 + (1 - 0.9397)(10{,}000 - 89) =$ **687**.

8 Double, Multiple & Sequential Sampling

slides 28–29

The advantage of a **double-sampling** plan over single sampling is that it may **reduce the total amount of inspection**. If the first (smaller) sample already gives a clear accept/reject, inspection cost is lower; a lot can also be rejected without completing the second sample.

9 MIL STD 105E

slides 39–54

MIL STD 105E is the most widely used attributes acceptance-sampling system (civilian equivalents: **ANSI/ASQC Z1.4**, **ISO 2859**). It provides **single, double and multiple** sampling, and is **AQL-oriented** (different AQLs for critical, major, minor defects).

Three inspection severities

Normal — used at the start. **Tightened** — when the supplier's recent quality has deteriorated (stricter acceptance). **Reduced** — when quality has been exceptionally good (smaller samples). Movement between them follows the **switching rules** (Fig 15.17).

Inspection levels

Sample size depends on **lot size** + **inspection level**. General levels: **I** (≈½ of II), **II = normal (most used)**, **III** (≈2× II). Plus four special levels **S-1…S-4** (very small samples).

Procedure

slide 43

1. Choose the AQL.
2. Choose the inspection level.
3. Determine the lot size.
4. Find the sample-size **code letter** (Table 15.4).
5. Choose the plan type (single / double / multiple).
6. Enter the appropriate master table to read the plan.
7. Determine the matching normal & reduced plans as required.

📊 **Example & Problem 15.22**

Example: N = 2,000, AQL = 0.65%, normal, level II → code letter **K** → single plan **n = 125, Ac (c) = 2, Re (r) = 3**.

Problem 15.22: N = 10,000, AQL = 0.10%, level II → code letter **L** → **n = 125, Ac = 0, Re = 1**.

⚠ **Discussion (15.4.3)**

MIL STD 105E is **AQL-oriented**; not all sample sizes occur (2, 3, 5, 8, 13, 20, 32, 50, ...); sample sizes are tied to lot sizes; the most common abuse is **failing to use the switching rules**.

📋 Formula Cheat Sheet

Quantity	Formula	Note
Prob. of acceptance (OC)	$Pa = \Sigma_{(d=0)}^{c} [n!/(d!(n-d)!)] p^d(1-p)^{n-d}$	binomial, Eq 15.2
Average outgoing quality	$AOQ = Pa \cdot p \cdot (N-n)/N \approx Pa \cdot p$	Eq 15.4
AOQL	max of the AOQ curve	worst average outgoing
Average total inspection	$ATI = n + (1-Pa)(N-n)$	Eq 15.6
Producer's risk	α = P(reject good lot)	at AQL
Consumer's risk	β = P(accept bad lot)	at LTPD

★ Quick Review

- Acceptance sampling **sentences lots** (accept/reject) — it does not estimate or improve quality.
- Plans: **single · double · multiple · sequential**; by attributes or variables.
- **AQL** (good, supplier's process) vs **LTPD** (poor, consumer protection); **AOQL** = max average outgoing defects.
- **OC curve:** Pa vs p; **α = reject good**, **β = accept bad**; larger n $\Longrightarrow$ steeper (more discriminating).
- Design single plan with the **binomial nomograph** from (AQL, $1-\alpha$) and (LTPD, β).
- **Rectifying:** AOQ = $Pa \cdot p \cdot (N-n)/N$; **ATI = $n + (1-Pa)(N-n)$**.
- **MIL STD 105E:** AQL-oriented; normal/tightened/reduced + switching rules; levels I/II/III; code letters → master tables.